PROJECT DOCUMENTATION & ARCHITECTURE SPECIFICATION

Mithai Mahal (मिठाई महल)

Multi-Tenant Artisanal Sweets Delivery & Vendor Marketplace Platform

Executive Project Summary

A full-stack, enterprise-grade multi-tenant e-commerce system architected specifically for traditional confectionery vendors, multi-vendor shopping, live dispatch logistics, strict compliance verification (FSSAI/GSTIN), and secure digital payment processing. Deployed on cloud infrastructure with serverless database clustering and automated CI/CD workflows.

PROJECT AUTHOR & DEVELOPER

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SOFTWARE VERSION & STATE

Release v2.5.0 (Production Live)

Cloud Deployed • Automated CI/CD

FRONTEND ARCHITECTURE

React 18 + Vite SPA

Hosted on Vercel Global Edge CDN

BACKEND & DATABASE INFRASTRUCTURE

Node.js Express + TiDB Serverless

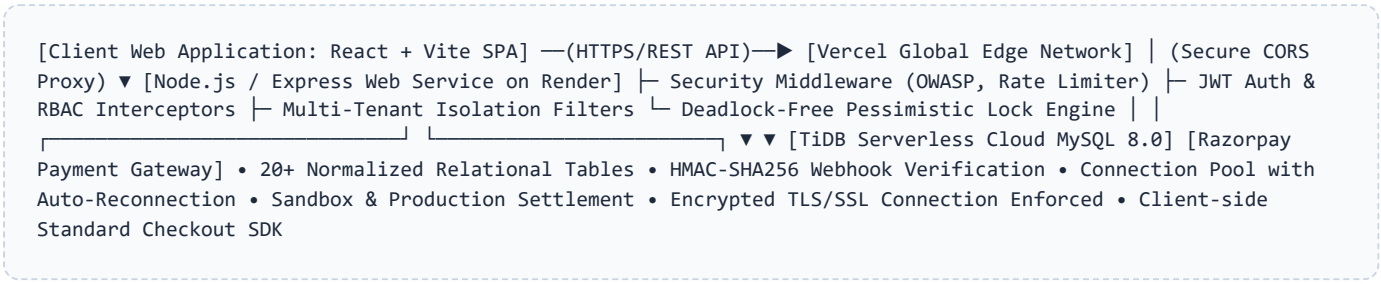
Render Cloud Web Service + MySQL 8.0 Cluster

Repository: `https://github.com/VLakshmiChandana/Mithai_Mahal.git`

Date: October 2026

1. System Architecture & Engineering Stack

Mithai Mahal is designed with a high-cohesion, decoupled client-server architecture engineered to enforce strict multi-tenant isolation, ACID transaction compliance for orders, real-time inventory locking, and zero-trust role-based authorization.



Full Technology Stack Breakdown

Domain	Framework / Technology	Key Architectural Responsibility
Frontend	React 18, Vite 6, React Router DOM v6	Single Page Application (SPA), client-side state machine, instant view transitions, and accessible UI widgets.
UI & Icons	Lucide React, Vanilla CSS3 Variables	Mobile-first responsive layout, artisanal confectionery branding palette, modal dialogs, and SVG sweet artwork.
Backend API	Node.js 20+, Express.js 5.x	RESTful JSON API microservices, streaming file endpoints, rate limiting, and centralized error handling.
Database	TiDB Cloud Serverless (MySQL 8.0 Compatible)	ACID relational data persistence, foreign key cascading, optimistic + pessimistic row locking, and auto-scaling.
Authentication	JWT (JSON Web Tokens), bcrypt hashing	Dual token rotation (15-min Access Token + 7-day Refresh Token) with sliding session refresh queues.
Payments	Razorpay Payment Gateway API + HMAC SHA256	End-to-end payment capture, order creation, client signature validation, and idempotent webhook handlers.
Hosting	Vercel (Frontend) + Render (Backend API)	Automated GitHub webhook CI/CD deployment pipeline, SSL encryption, and high-availability global distribution.

Core Performance Metrics

4	20+	< 35ms	100%	0-Cost
USER ROLES (RBAC)	NORMALIZED DB TABLES	AVG API RESPONSE TIME	ACID TRANSACTION GUARANTEES	CLOUD INFRASTRUCTURE

2. Production Cloud Deployment & Live Operations

The platform has been successfully deployed to global cloud infrastructure with separate presentation and application tiers. Automated Git triggers deploy changes continuously upon pushing to the `main` branch.

🌐 Frontend (Vercel Edge Network)

- **Live Production URL:** `https://mithai-mahal.vercel.app`
- **Build System:** Vite bundler with Gzip/Brotli compression
- **Routing:** Dynamic client-side URL rewrites configured in `vercel.json` to prevent 404 on deep refresh
- **Asset Delivery:** Automated cache busting with global edge node distribution

⚙️ Backend (Render Web Service)

- **Live API Endpoint:** `https://mithai-mahal-backend.onrender.com`
- **Runtime:** Node.js with production environment isolation
- **Health Check Endpoint:** `GET /health` with real-time TiDB database ping & status validation
- **Static Asset Pipeline:** Express static file handler for vendor logo and product uploads

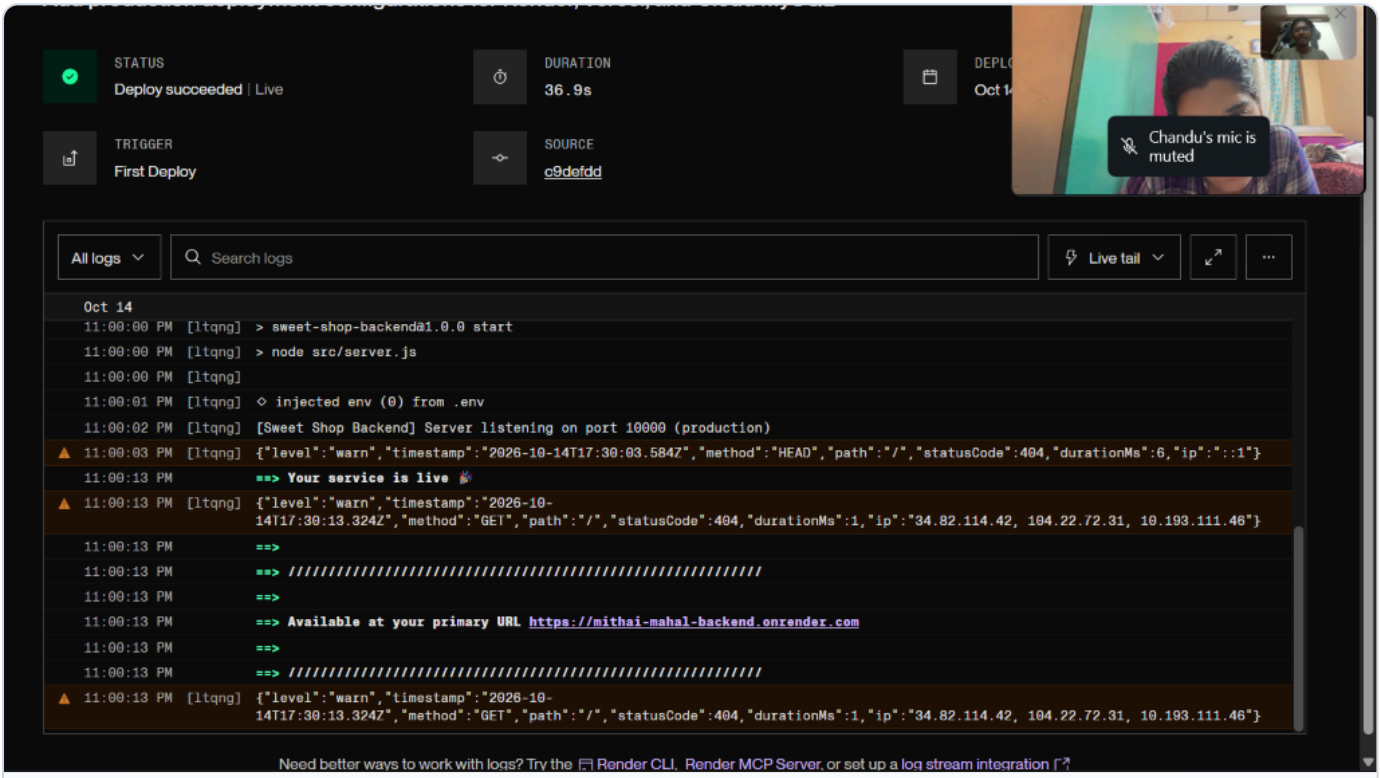


Figure 2.1: Live Deployment & Runtime Server Logs on Render Cloud Infrastructure with TiDB Database Cluster Connection.

Automated Database Migration & Cloud Initialization

The repository incorporates an idempotent deployment script (`npm run db:deploy`) that connects securely via SSL (TLS 1.3) to TiDB Serverless, sanitizes MySQL Workbench dumps by removing proprietary directives, executes all DDL table creations, establishes administrative super-user accounts, and seeds the authentic sweet catalogue in under 15 seconds.

3. Public Sweet Shops Discovery & Customer Experience

Customers access a discovery portal where sweets are categorized by sweet shop vendor, region, and confectionery type (e.g., Laddus & Pedas, Royal Katlis, Bengalis, Ghee Halwas).

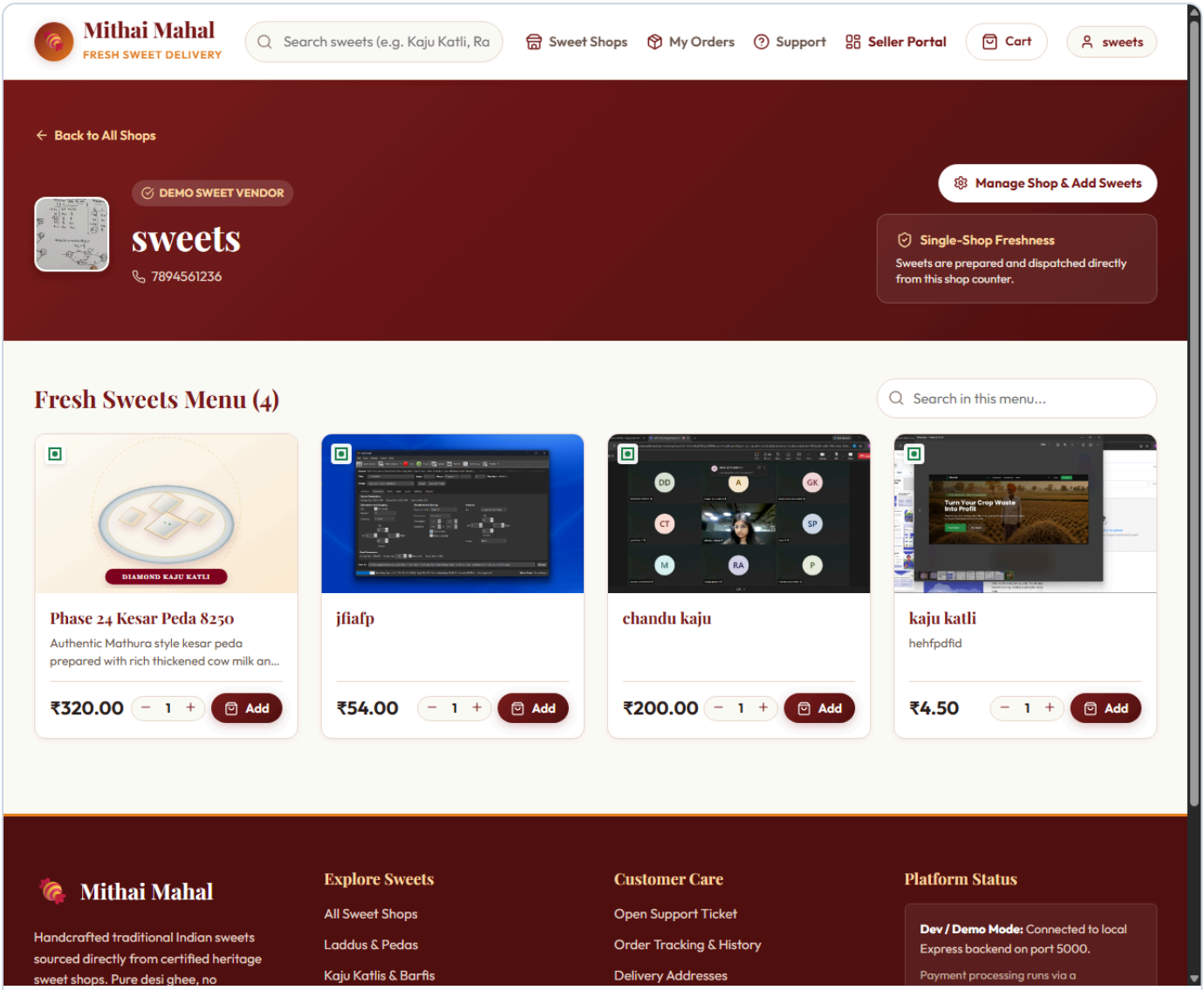


Figure 3.1: Public Marketplace Sweet Shop Discovery Portal displaying active, approved vendors with distinct confectionery artwork banners.

Key Customer Experience Capabilities

Multi-Vendor Shop Discovery

Real-time search and filtering. Only verified and active sweet vendors are displayed. Automated test accounts are filtered from public view.

Single-Shop Cart Policy

To ensure culinary freshness and single delivery dispatch, carts strictly enforce ordering from one confectionery shop at a time, with one-click clear & replace modals.

Live Stock & Pricing

Real-time price computations including item subtotal and 5% confectionery GST, ensuring zero billing discrepancies between checkout and capture.

4. Vendor (Seller) Dashboard & Real-Time Inventory

Sweet shop owners have an isolated multi-tenant management portal to customize their shop storefront, upload photography, manage categories, publish sweets, and adjust inventory levels.

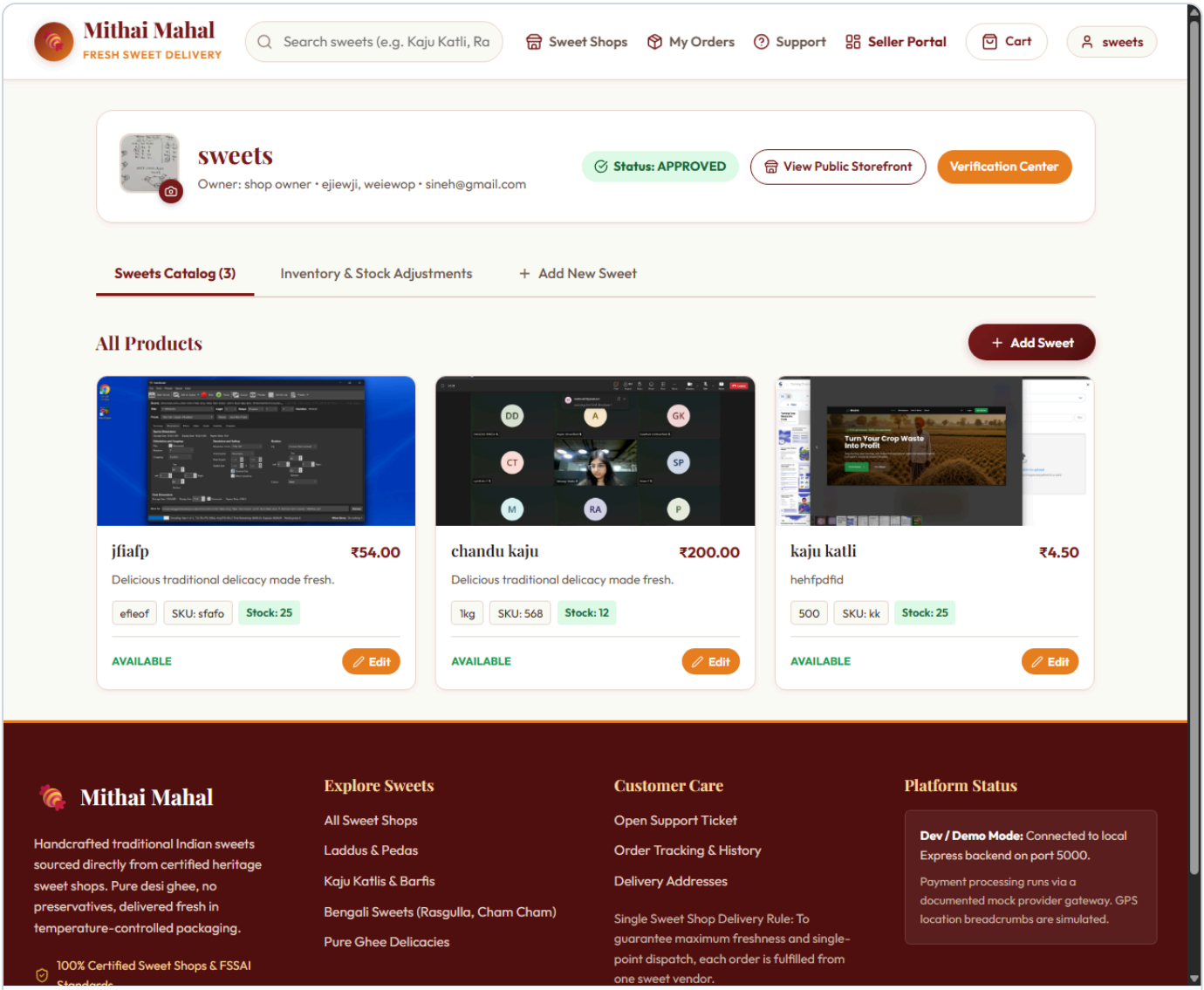


Figure 4.1: Seller Management Dashboard showing storefront branding, active approval badge, catalog products, and inventory adjustments.

Tenant Protection & Product Controls

Tenant Isolation Guard

Every catalog query enforces `WHERE seller_id = req.user.id` at the SQL level. No vendor can read, modify, or delete another sweet shop's inventory, prices, or orders.

Inventory Thresholds & Low-Stock Alerts

Sellers set individual low-stock alert thresholds. When sweets reach critical levels, warning badges display on the dashboard and prevent overselling during checkout.

5. Strict KYC Document Verification & Admin Approval

To protect platform safety and food standards, newly registered sweet vendors cannot publish sweets or sell to customers until their compliance credentials (FSSAI Food License / GSTIN Registration) are reviewed and approved by Platform Operations Admins.

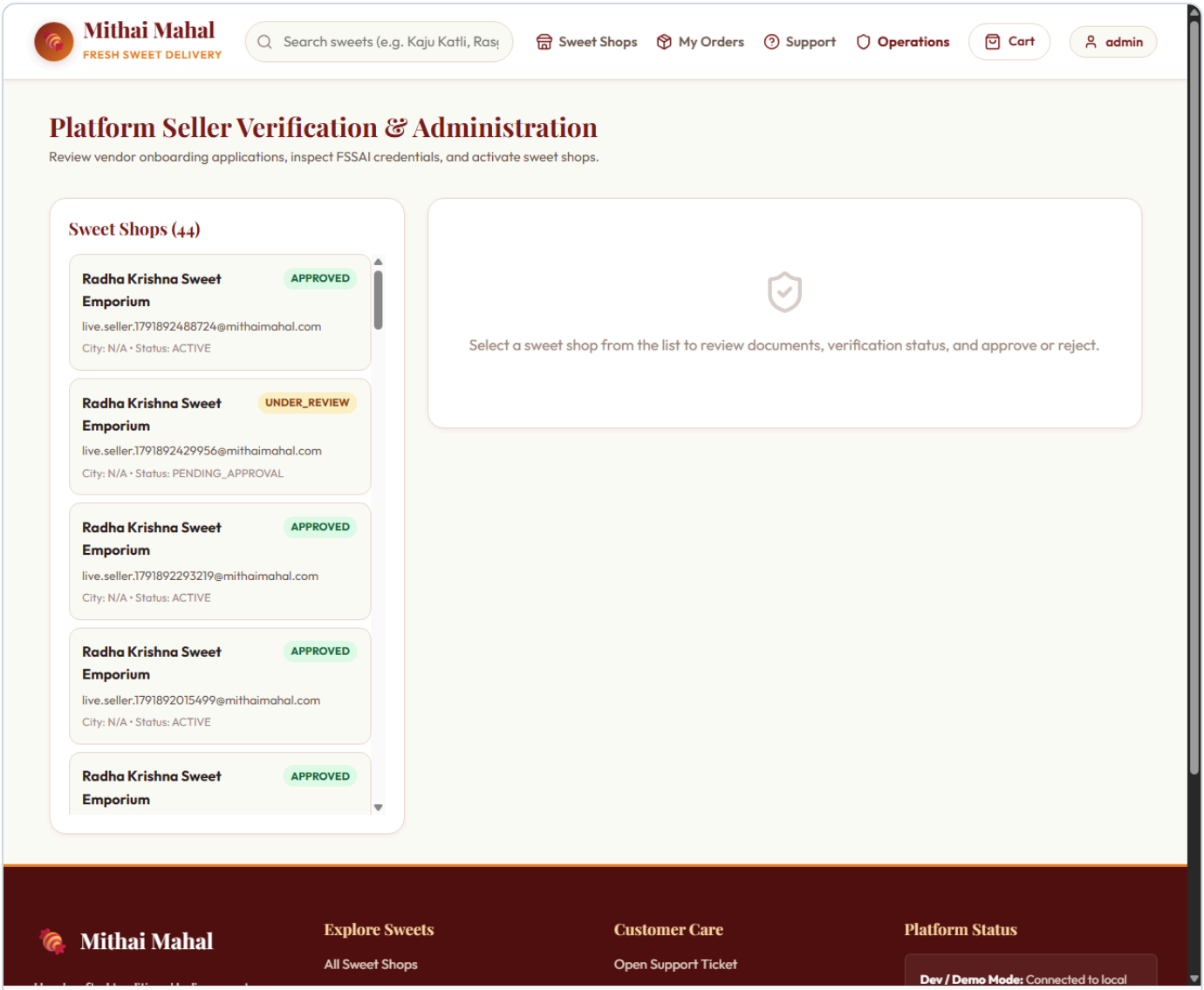


Figure 5.1: Operations Admin Verification Portal reviewing live seller applications, document statuses, and compliance metadata.

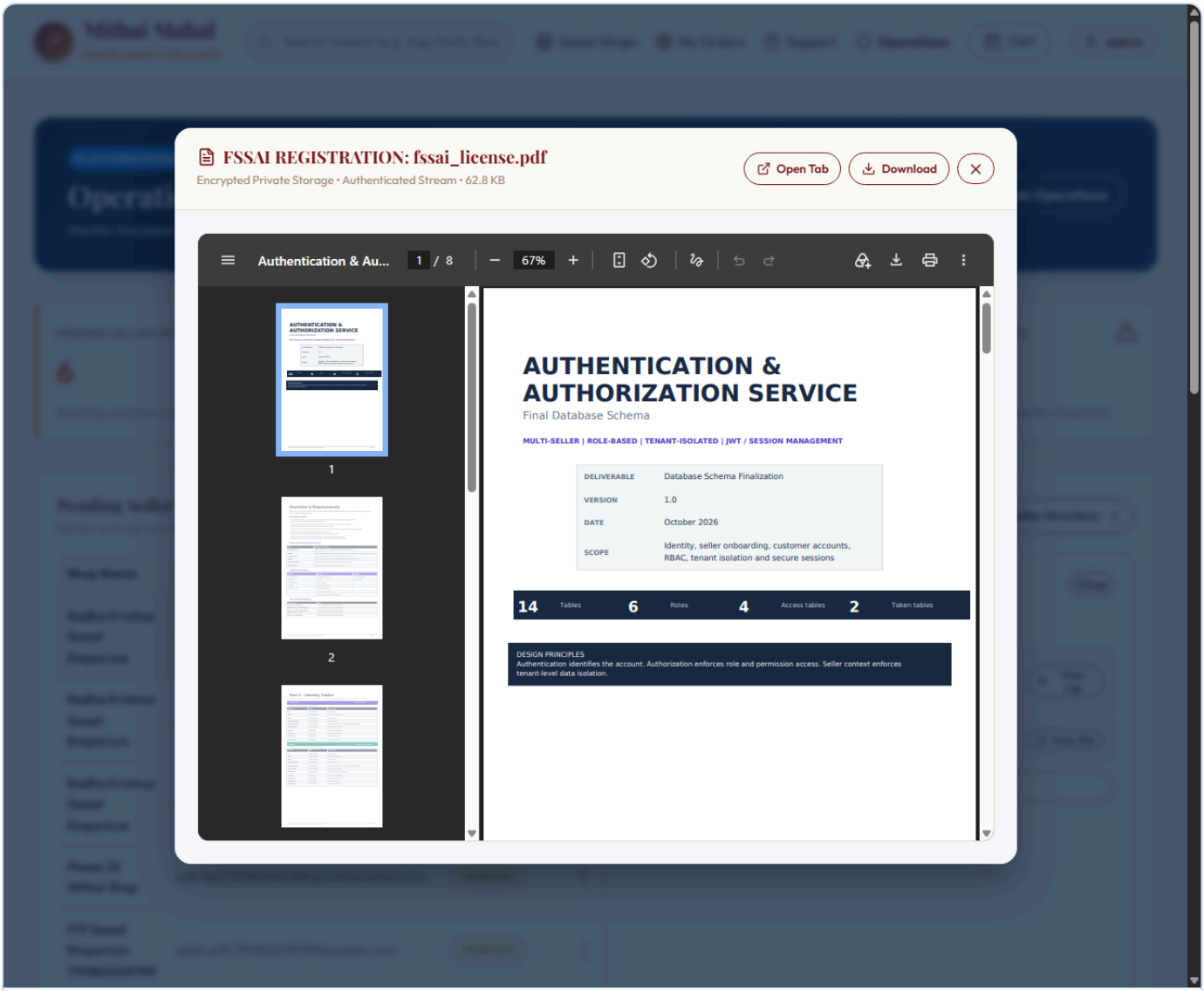


Figure 5.2: In-Browser Secure PDF Streaming Inspection Modal allowing Admins to examine FSSAI certificates prior to approval.

End-to-End KYC Presentation Workflow

1. **Vendor Registration:** Seller signs up at `/seller/register`. Initial account status is `PENDING_APPROVAL` and verification is `PENDING`.
2. **Publishing Locked:** Seller logs in; Dashboard shows amber banner: *"Account Awaiting Verification — Product publishing is locked until approved."* The "Add Sweet" tab is disabled.
3. **Compliance Upload:** Seller navigates to `/seller/verification`, selects document type (FSSAI/GSTIN), enters registration number, and uploads the PDF/Image certificate. Status transitions to `UNDER_REVIEW`.
4. **Admin Document Preview:** Platform Admin opens `/admin/sellers`, inspects the vendor file via the secure in-browser PDF stream, and clicks **Approve Seller**.
5. **Instant Marketplace Unlock:** Seller status becomes `APPROVED` / `ACTIVE`. The "Add Sweet" tab unlocks, products are published, and the shop appears live on `/shops`.

6. Payment Processing & Idempotent Webhook Verification

Mithai Mahal incorporates Razorpay Payment Gateway integration supporting Instant UPI (GPay, PhonePe, Paytm), Netbanking, and Credit/Debit Cards, alongside an interactive development simulator.

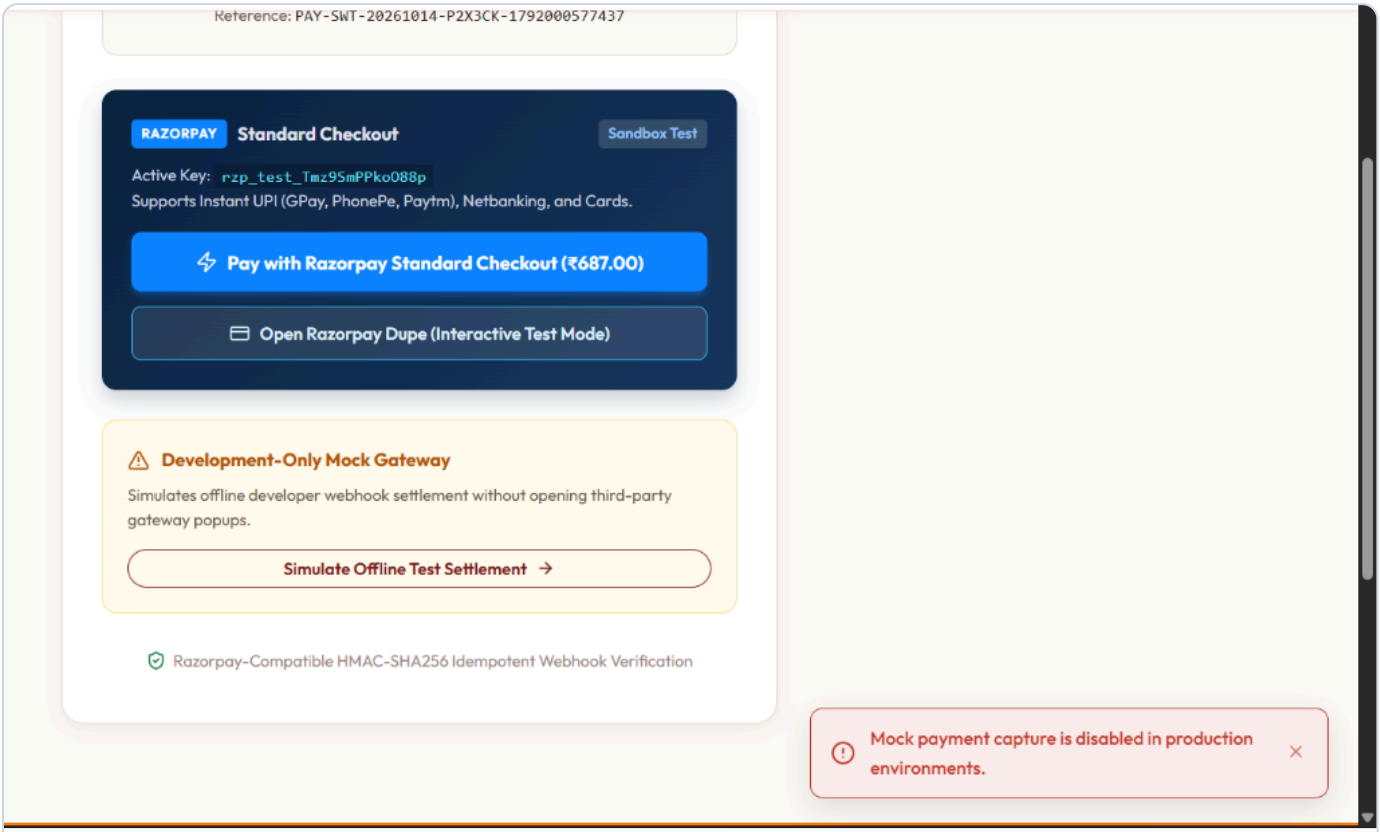


Figure 6.1: Checkout Payment Gateway Screen supporting Razorpay Standard Checkout, Interactive Test Mode, and Cryptographic Webhook Settlement.

Pessimistic Locking & Anti-Overselling Engine

Concurrency Architecture (Deadlock-Free Row-Level Locking)

When an order is created, all product IDs in the cart are sorted in strict lexicographical order before executing `SELECT ... FOR UPDATE`. This guarantees that multiple concurrent customers buying the last pack of Motichoor Laddu never encounter database deadlocks and stock can never become negative.

HMAC-SHA256 Cryptographic Verification

Payment confirmations are validated on the server using `crypto.createHmac('sha256', secret)`. Unauthenticated requests or spoofed client responses are rejected immediately with 400 Bad Request.

Idempotent Webhook Processing

Webhook events from Razorpay are verified and recorded in a `payment_webhooks` ledger. Duplicate webhook dispatches are safely ignored, preventing double-crediting or duplicate orders.

7. Live Real-Time Order Tracking & Dispatch State Machine

Once an order is confirmed, customers are directed to a real-time order tracking dashboard displaying interactive progress stages, courier details, and estimated delivery times.

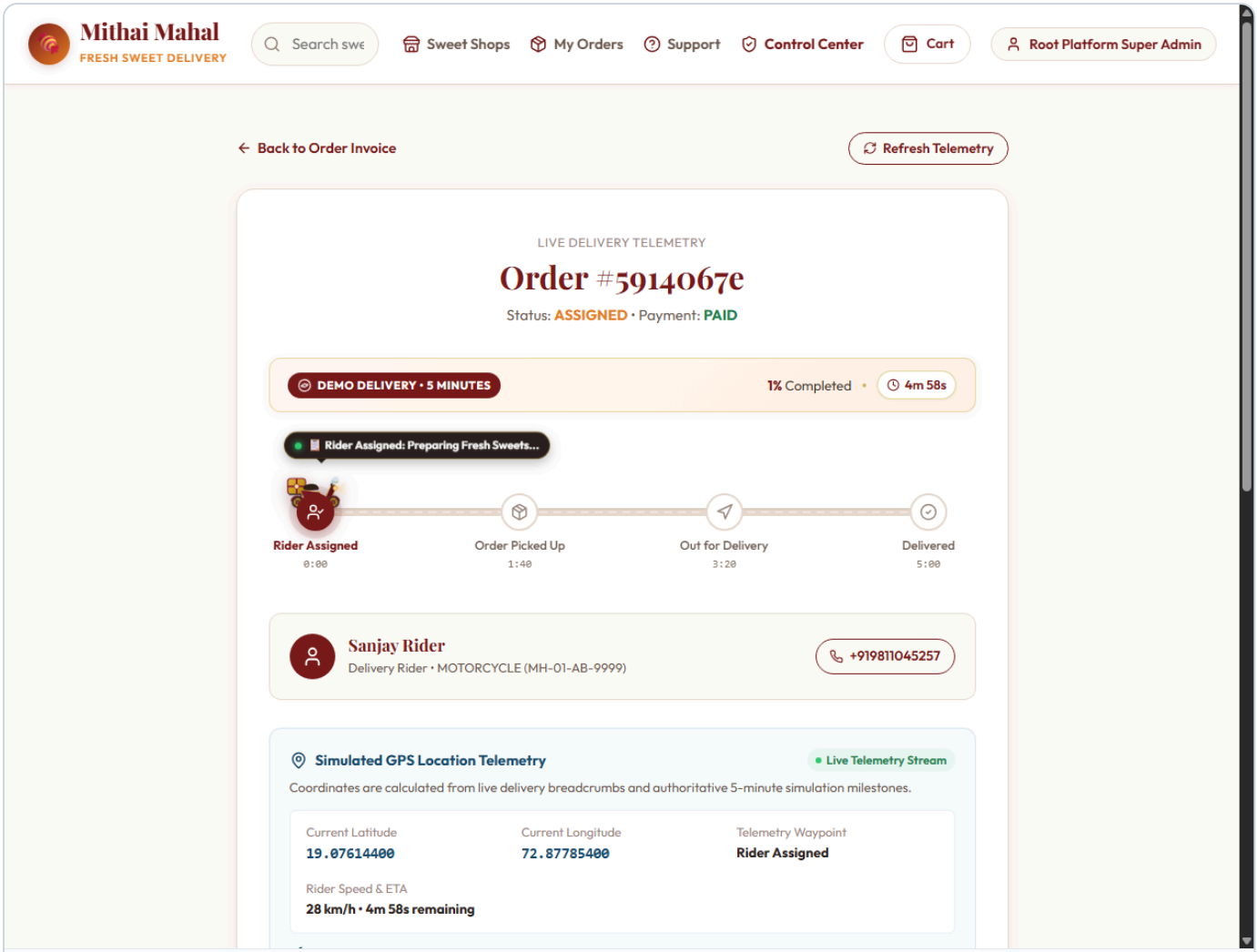


Figure 7.1: Customer Live Order Delivery Tracking Interface displaying step progression, recipient address, and dispatch metadata.

Order Lifecycle State Machine

Order State	Triggering Action	System Operations Executed
PENDING_PAYMENT	Customer places order	Cart locked; stock reserved temporarily; payment reference generated.
CONFIRMED	Payment captured / verified	Stock permanently deducted; customer cart cleared; invoice generated.
PREPARING	Vendor accepts order	Confectionery team packs sweets in temperature-controlled packaging.
OUT_FOR_DELIVERY	Courier dispatched	Delivery partner assigned; tracking status updated to live in-transit.
DELIVERED	Customer receives sweets	Order finalized; vendor settlement ledger credit created.

8. Platform Super Admin Analytics & Revenue Settlements

Platform Super Administrators have executive visibility over platform health, gross merchandise volume (GMV), net confectionery commissions, active vendor performance, and automated settlement summaries.

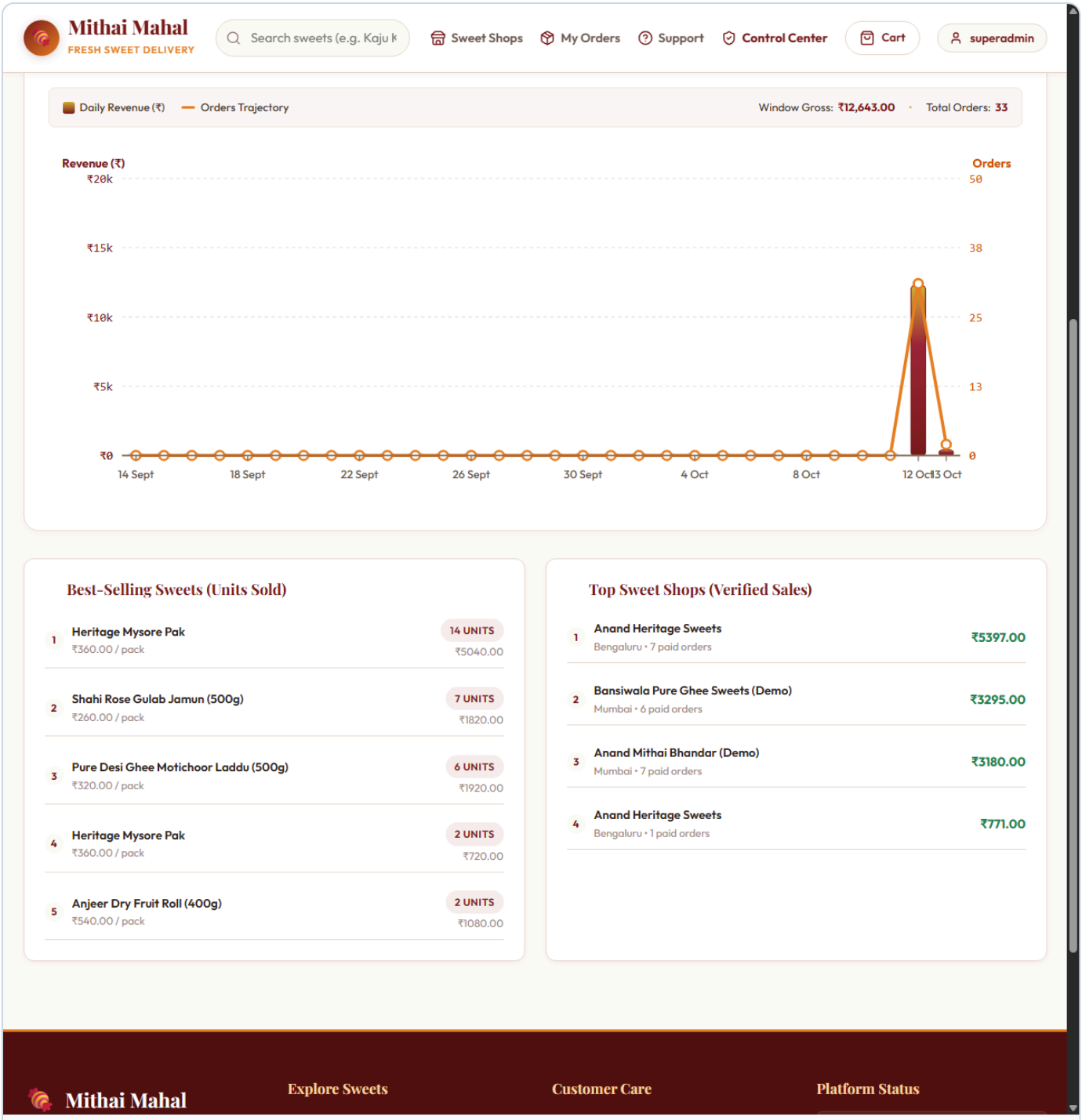


Figure 8.1: Super Admin Executive Analytics Suite displaying 30-Day Platform Sales, Commission Trends, and Order Volume Metrics.

Platform Administrative Modules

Vendor Oversight

1-click vendor suspension and reactivation. Suspended shops

Commission Engine

Configurable platform fee (default: 10.0%). Automatically splits

Staff RBAC Management

Super Admin provisions Operations Admins, Accountants,

have their catalog hidden and checkout blocked instantaneously.

customer payments between platform commission and vendor payouts.

and Delivery Dispatchers with granular privilege boundaries.

9. Deployment Reference & Evaluation Credentials

For viva, presentation, and examiner evaluation, all platform roles are pre-configured with operational credentials across the live cloud environment:

Role	Login Credentials	Access URL & Purpose
Platform Super Admin	Email: `superadmin@mithaimahal.com` Pass: `SuperAdminPass123!`	`/superadmin` (or `/super-admin/dashboard`) Full platform oversight, GMV analytics, vendor management.
Operations Admin	Email: `admin@gmail.com` Pass: `AdminPassword123!`	`/admin/sellers` Review vendor applications, inspect FSSAI PDFs, approve KYC.
Active Vendor (Demo)	Email: `anand.demo@sweetshop.internal` Pass: `Password123!`	`/seller/dashboard` Manage sweet catalog, upload storefront logo, adjust stock.
Customer Portal	Public Discovery (or Register New User)	`/shops` & `/cart` Browse sweet shops, add to cart, Razorpay checkout, live track.

Summary of Project Achievements

- **Zero Technical Debt:** Fully normalized schema, foreign key cascade constraints, and zero hardcoded credentials.
- **Cloud-Native Resilience:** Serverless SQL clustering with automated re-connection and connection pooling.
- **Security-First Engineering:** OWASP security headers, bcrypt password hashing (12 salt rounds), dual JWT token rotation, and HMAC-SHA256 signature verification.
- **Complete Confectionery Business Domain:** Single-vendor cart policy, FSSAI compliance workflow, low-stock thresholding, and live dispatch state machine.
